

Codeless Content Creator System: Anyone Can Make Their Own Mixed Reality Content Without Relying on Software Developer Tools

Suhyeon Kim
School of Computing, Gachon
University
Republic of Korea
kih629@gachon.ac.kr

Dokyung Lee
School of Computing, Gachon
University
Republic of Korea
cads225@gachon.ac.kr

Jaejun Park
School of Computing, Gachon
University
Republic of Korea
3636qq@naver.com

Myungji Song
School of Computing, Gachon
University
Republic of Korea
qqqgina@gachon.ac.kr

Younhyun Jung
School of Computing, Gachon
University
Republic of Korea
younhyun.jung@gachon.ac.kr

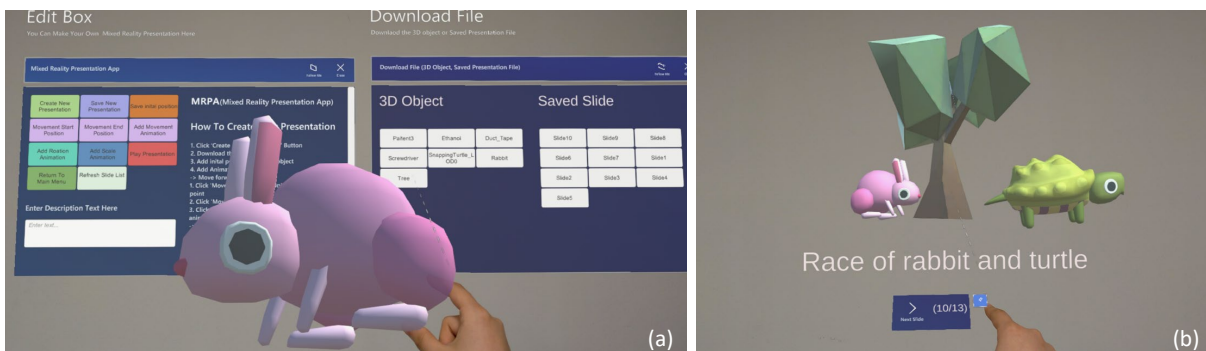


Figure 1: An example of (a) Edit mode and (b) Play mode.

ABSTRACT

Mixed Reality (MR) techniques begin to penetrate our daily life. Due to the features of MR techniques to seamlessly intermix a user's real environment with useful virtual contents in a natural manner, their potential applications range from information visualization and sharing and remote work collaboration to education and training in various fields. However, due to the steep learning curve in content creation and development, it is not trivial for novice users to create MR contents by themselves. In this work, we propose "Codeless Content Creator System", an MR system that helps novice users rapidly and conveniently create MR contents using MR devices rather than relying on complicated software development tools such as Unity. A simple user study found that our system is useful and convenient for the MR contents creation tasks.

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1 INTRODUCTION

Mixed Reality (MR) is a useful working space where a user's real environment can be seamlessly intermixed with relevant virtual contents. The studies by Melanie et al. [Maas and Hughes 2020] and Tang et al. [Tang et al. 2020] found that when contents/information are delivered through and interacted with MR technology, a user can easily find interest and immersion and increase their understanding of information. As such, MR technologies can be effective, in particular, for information visualization and sharing, remote work collaboration, and education and training in various fields [Gerup et al. 2020] [Jurascchek et al. 2018].

One of the barriers to prevent the wider adoption of MR usage is technological complexity in content creation; novice users who wish to utilize MR quickly and easily should rely on dedicated MR development tools. In the dominant technological ecosystem of Microsoft MR products, the understanding of and the experiences on low-level programming language such as C++ and C# and powerful

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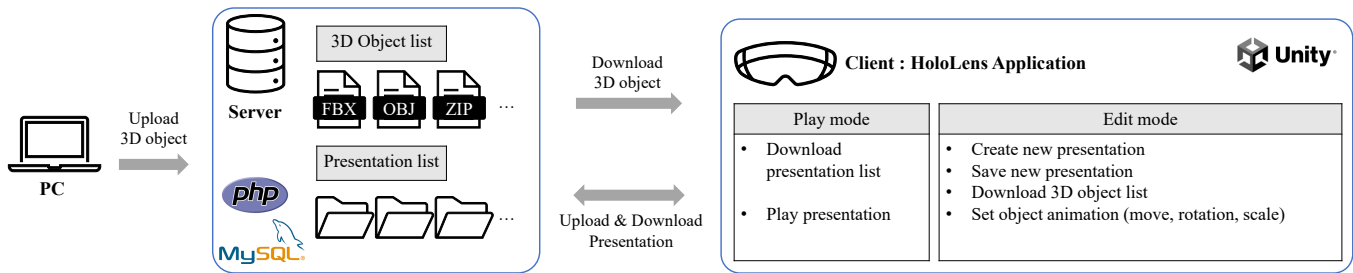


Figure 2: Codeless Content Creator System Overview

but complicated Unity platform must be required even for a simple MR content creation.

In the study, we propose “Codeless Content Creator System”, an MR system that enables a user to develop and deploy their own MR content without relying on the complicated software development tools (as shown in Figure 1). Using our system, a user can build a series of scenes with 3D objects and different animation effects and text descriptions. The resultant MR contents are then visualized and interacted in real-time. We implemented our system based on HoloLens 2 MR headset and evaluated its usability through the informal feedbacks from five novice users.

2 IMPLEMENTATION

Our system is based on a server-client model (as shown in Figure 2) : (i) the server that stores and manages user-created contents with 3D objects and action scripts; and (ii) Microsoft HoloLens 2 client for the content creation and display.

Server was implemented using PHP and MySQL. Using MySQL, 3D objects and presentations are properly managed, and php is used to communicate between the Unity application and MySQL. We support various formats, including FBX, OBJ, and ZIP, to store 3D objects that can be used in user-created contents. Users can upload or delete desired 3D objects using a PC web browser.

Our client system was implemented using Unity and C#. The application has two modes. 1) ‘Edit mode’ when a user creates MR contents and 2) ‘Play mode’ when the user plays them. The user interface for the Edit mode is shown in Figure 1 (a). A user can select 3D objects to be included in their contents from the list that is dynamically called through the server. The chosen 3D objects are downloaded using TriLib Assets and generated as a GameObject for use in the MR contents. For the user to be able to freely interact with 3D objects using hand gesture, Near Interaction and Object Manipulate Script are added along with a Box Collider that is built based on the size of the GameObject.

User can add description text, and assign animations to each of the objects. Animations include movement, scale-up, and rotation. Use the grab event trigger to add animations to the most recent object the user has grabbed. The target object name, animation type, and target position are bound in order to record animations. All actions performed by the user are scripted and constitute a presentation, which is then transferred and saved on the server.

During the Play mode, the user is given the content list from the server. In Figure 1 (b), the Play mode with the chosen MR story, ‘The Tortoise and the Hare’, is illustrated. It includes a series of

story contents. When the “Next Slide” button is pressed, the actions created in the Edit mode are executed in sequence. Here, we can see Tortoise overtaking the hare.

3 USER STUDY / USER EXPERIENCE

A simple user study was conducted on how easily MR contents can be created when using our codeless content creator system. We recruited 5 computer science major bachelor students (3 male, 2 female) who have understanding of MR technologies. At first, the experimenters learned how to operate the HoloLens2 simply. They were then guided on how to use our system by showing a demo video about the step-by-step process of creating a MR content for the story, ‘The Tortoise and the Hare’, using the Edit mode. After that, the experimenter wore HoloLens 2 and created their own ‘The Tortoise and the Hare’ content. They confirmed their final content through the play mode and wrote a subjective evaluation to see if our system has any values.

All the experimenters responded that our system is useful when content creation is made in a MR environment and that novice users who are not developers also can create simple MR contents through our proposed system in a convenient and rapid manner. The two experimenters answered that “the applicability and the utility in various fields are expected.”

4 CONCLUSION AND FUTURE WORK

In this work, we present “Codeless Content Creator System”, an system that assists novice users in creating and experiencing MR contents in a rapid and user-friendly manner. The simple user study found some values in the usability and utilities of our system. As a future work, we plan to improve functionalities and user experiences by adding a voice recording function, multi-user sharing using Mesh and Photon, and simultaneous interaction with multiple objects.

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